

Statistical Methods in Experimental Particle Physics

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Search for the lepton-flavour violating $\tau^+ \rightarrow \phi\mu^+$ decay

Introduction

Searches for charged lepton-flavour violating decays provide stringent tests of the Standard Model and of its possible extensions. In the Standard Model with massive neutrinos, such processes are allowed only at rates far below any foreseeable experimental sensitivity; an observation of the $\tau^+ \rightarrow \phi\mu^+$ decay at current experiments, such as LHCb or Belle II, would therefore constitute unambiguous evidence for New Physics. The current best frequentist upper limit is set by Belle [1], which reports

$$\mathcal{B}(\tau^+ \rightarrow \phi\mu^+) < 8.4 \times 10^{-8} \quad \text{at 90\% CL.}$$

In the following simulated experiment a search for this decay process is proposed in an environment emulating the conditions of the LHCb experiment at the Large Hadron Collider. The experimenter is asked to provide the best estimate of this branching fraction, or the corresponding confidence interval/upper limit at 90% CL, assessing both statistical and systematic uncertainties. An interesting reading on the subject can be found in Ref. [2].

Experimental apparatus and data

Simulated data are produced using a simplified parametric simulation of the LHCb experiment [5]. The curvature of individual particles in the xz plane is smeared, as well as the position of the primary vertex (PV) and decay vertex (DV). The LHCb coordinate system is a right-handed system centred in the nominal pp collision point $(0, 0, 0)$, with the z axis pointing along the beam direction towards the downstream detectors, the y axis pointing vertically upwards, and the x axis pointing in the horizontal direction, outside the centre of the LHC. Both the data and simulated

Sample	Content	N_{gen}
<code>tree_data.root</code>	reconstructed candidates in the $\phi(\rightarrow K^+ K^-)\mu^+$ final state	—
<code>tree_DSPLUS_TauNu_mc_1M.root</code>	$D_s^+ \rightarrow \tau^+ \nu_\tau$, with $\tau^+ \rightarrow \phi(\rightarrow K^+ K^-)\mu^+$	10^6
<code>tree_DSPLUS_PhiPi_mc_50M.root</code>	$D_s^+ \rightarrow \phi(\rightarrow K^+ K^-)\pi^+$	50×10^6
<code>tree_DPLUS_PhiPi_mc_50M.root</code>	$D^+ \rightarrow \phi(\rightarrow K^+ K^-)\pi^+$	50×10^6
<code>tree_DSPLUS_PhiMuNu_mc_5M.root</code>	$D_s^+ \rightarrow \phi(\rightarrow K^+ K^-)\mu^+ \nu_\mu$	5×10^6

Table 1: Data and simulated samples available for the experiment.

samples available for the experiment are listed in Tab. 1. They are collected or generated as in a real experiment, namely within the geometrical acceptance of the detector and after applying the trigger requirements reported in Tab. 2. Only prompt production of $D_{(s)}^+$ mesons in proton-proton collisions is considered in this experiment, with no contamination from b -hadron decays or from promptly produced τ leptons. To simplify both the experiment and the data analysis, τ^+ leptons are produced only through $D_s^+ \rightarrow \tau^+ \nu_\tau$ decays. The Monte Carlo samples are generated using the known PDG values for particle masses and lifetimes [3].

Observable	Requirement	Units
$p_T(h_{1,2})$	> 0.450	GeV/c
$p(h_{1,2})$	> 3.0	GeV/c
$\eta(h_{1,2})$	$2.0 < \eta < 4.2$	–
$IP(h_{1,2})$	> 60	μm
$p_T(h_3)$	> 0.350	GeV/c
$p(h_3)$	> 3.0	GeV/c
$\eta(h_3)$	$2.0 < \eta < 4.2$	–
$IP(h_3)$	> 90	μm
$\text{MuonID}(h_3)$	$= 1$	–
$p_T(\phi)$	> 0.9	GeV/c
$p_T(\phi h_3)$	> 2.5	GeV/c
$ct(\phi h_3)$	> 0.25	ps
$m(\phi h_3)$	$1.6 < M < 2.1$	GeV/c ²

Table 2: Trigger and offline requirements applied to the reconstructed $\tau^+ \rightarrow \phi(\rightarrow h_1^- h_2^+) h_3^+$ candidates. In the signal hypothesis, h_1 and h_2 are kaon candidates and h_3 is the muon candidate.

Stored observables and basic definitions

The following observables are stored in ROOT TTree objects, where the convention $D^+ \rightarrow M^0 h_3^+ \rightarrow [h_1^- h_2^+] h_3^+$ is used for the decay of an intermediate neutral state M^0 from the τ^+ lepton or $D_{(s)}^+$ meson. In this exercise M^0 stands for the ϕ meson, which is reconstructed in the $K^+ K^-$ final state. The prefix D denotes the reconstructed three-body parent candidate. For the signal sample this candidate corresponds to the reconstructed τ^+ , while for the background samples it corresponds to a reconstructed charm-hadron candidate.

Event identifiers

- **id** – Only for Monte Carlo decays. For data $id \equiv 0$. For MC $id \equiv 34$ for $D^+ \rightarrow \phi \pi^+$ decays, and $id \equiv 41, 42, 43, 44$ for $D_s^+ \rightarrow \phi \pi^+$, $D_s^+ \rightarrow \phi \mu^+ \nu_\mu$, $D_s^+ \rightarrow \phi e^+ \nu_e$, $D_s^+ \rightarrow \tau^+ \nu_\tau$ decays, respectively.
- **event** – This is empty (999).

Final-state tracks

- **h1:px,py,pz,pt,p,eta,IP,charge,MuonID** – Momentum, pseudo-rapidity, impact parameter, charge, and Muon ID of the h_1^- particle.
- **h1:px_true,py_true,pz_true** – True values of the momentum components of the h_1^- particle. Only available in the Monte Carlo file.
- **h2:px,py,pz,pt,p,eta,IP,charge,MuonID** – Momentum, pseudo-rapidity, impact parameter, charge, and Muon ID of the h_2^+ particle.
- **h2:px_true,py_true,pz_true** – True values of the momentum components of the h_2^+ particle. Only available in the Monte Carlo file.
- **h3:px,py,pt,pz,p,eta,IP,charge,MuonID** – Momentum, pseudo-rapidity, impact parameter, charge, and Muon ID of the h_3^+ particle.

Primary and decay vertices

- **PV:x,y,z** – Coordinates of the position of the primary vertex.

- **PV:x_true,y_true,z_true** – True values of the coordinates of the position of the primary vertex. Only available in the Monte Carlo file.
- **DV:x,y,z** – Coordinates of the position of the decay vertex of the M^0 meson.
- **DV:x_true,y_true,z_true** – True values of the coordinates of the position of the decay vertex of the M^0 meson. Only available in the Monte Carlo file.

Intermediate M^0 candidate

- **M0:px,py,pt,pz,p,eta** – Momentum and pseudo-rapidity of the M^0 meson.
- **M0:FDx,FDy,FDt,FDz,FD** – Flight distance of the M^0 meson with respect to the PV.
- **M0:time** – Proper decay time of the M^0 meson.
- **M0:time_true** – True proper decay time of the M^0 meson. Only available in the Monte Carlo file.
- **M0:IP** – Impact parameter of the M^0 meson.
- **M0:Mpipi,MKK,MKpi,MpiK** – Invariant masses calculated with different mass hypotheses.

Three-body parent candidate

- **D:Vx,Vy,Vz** – Coordinates of the position of the vertex of the three-body parent candidate.
- **D:Vx_true,Vy_true,Vz_true** – True values of the coordinates of the position of the vertex of the three-body parent candidate. Only available in the Monte Carlo file.
- **D:px,py,pt,pz,p,eta** – Momentum and pseudo-rapidity of the three-body parent candidate.
- **D:FDx,FDy,FDt,FDz,FD** – Flight distance of the three-body parent candidate with respect to the PV.
- **D:time** – Proper decay time of the three-body parent candidate. For the signal sample this refers to the τ lepton; therefore, the τ mass from the PDG is used.
- **D:time_true** – True proper decay time of the three-body parent candidate. Only available in the Monte Carlo file.
- **D:IP** – Impact parameter of the three-body parent candidate.
- **D:M** – Invariant mass calculated with the $M^0 \equiv \phi$ and $h_3 = \mu$ mass hypotheses.

Momenta are measured in GeV/c, masses in GeV/c², energies in GeV, lengths in metres, times in seconds, and angles in radians.

Objectives and general analysis strategy

The goal of this experiment is to search for the rare $\tau^+ \rightarrow \phi\mu^+$ decay process and to determine the branching fraction

$$\mathcal{B}(\tau^+ \rightarrow \phi\mu^+).$$

The student is asked to provide the best estimate of this quantity and to report either a confidence interval or an upper limit at 90% CL, assigning the statistical uncertainty and discussing the dominant systematic uncertainties. Equivalently, one may want also to measure the ratio

$$R_\tau = \frac{\mathcal{B}(D_s^+ \rightarrow \tau^+\nu_\tau)\mathcal{B}(\tau^+ \rightarrow \phi\mu^+)}{\mathcal{B}(D_s^+ \rightarrow \phi\mu^+\nu_\mu)},$$

which is free from external inputs.

A standard practice is to determine the signal yield relative to that of another decay mode, referred to as the normalisation channel, which shares similar features with the signal. The semileptonic $D_s^+ \rightarrow \phi \mu^+ \nu_\mu$ decay has the same visible final state as the signal and is therefore well suited to be used as the normalisation channel. In this case, the expected yield ratio can be written as

$$\frac{N_{\text{sig}}}{N_{\phi\mu\nu}} = \frac{\varepsilon_{\text{sig}}}{\varepsilon_{\phi\mu\nu}} \frac{\mathcal{B}(D_s^+ \rightarrow \tau^+ \nu_\tau) \mathcal{B}(\tau^+ \rightarrow \phi \mu^+)}{\mathcal{B}(D_s^+ \rightarrow \phi \mu^+ \nu_\mu)},$$

where $N_{\text{sig}}/N_{\phi\mu\nu}$ is the measured ratio of the signal yield to the semileptonic normalisation yield, while $\varepsilon_{\text{sig}}/\varepsilon_{\phi\mu\nu}$ is the corresponding reconstruction-efficiency ratio, determined using Monte Carlo samples.

As a validation of the full analysis procedure, it is strongly recommended to also measure the following branching-fraction ratio:

$$R_{\phi\pi} = \frac{\sigma(pp \rightarrow D^+ X) \cdot \mathcal{B}(D^+ \rightarrow \phi \pi^+)}{\sigma(pp \rightarrow D_s^+ X) \cdot \mathcal{B}(D_s^+ \rightarrow \phi \pi^+)}.$$

Experimentally, this ratio can be inferred from the measured yields of the two hadronic peaking components as

$$R_{\phi\pi} = \frac{N_{D^+ \rightarrow \phi \pi^+}}{N_{D_s^+ \rightarrow \phi \pi^+}} \frac{\varepsilon_{D_s^+ \rightarrow \phi \pi^+}}{\varepsilon_{D^+ \rightarrow \phi \pi^+}},$$

thereby providing a fully data-driven validation of the reliability of the simulation in reproducing the properties of the two peaking misidentified backgrounds, $D^+ \rightarrow \phi \pi^+$ and $D_s^+ \rightarrow \phi \pi^+$.

In order to perform the measurements it is useful to know the following values [3]:

- $\mathcal{B}(D^+ \rightarrow \phi \pi^+ \rightarrow [K^+ K^-] \pi^+) = (2.69_{-0.08}^{+0.07}) \times 10^{-3}$,
- $\mathcal{B}(D_s^+ \rightarrow \phi \pi^+ \rightarrow [K^+ K^-] \pi^+) = (2.25 \pm 0.05) \times 10^{-2}$,
- $\mathcal{B}(D_s^+ \rightarrow \phi \mu^+ \nu_\mu) = (2.24 \pm 0.11) \times 10^{-2}$,
- $\mathcal{B}(D_s^+ \rightarrow \tau^+ \nu_\tau) = (5.39 \pm 0.09) \times 10^{-2}$,
- $\mathcal{B}(\phi \rightarrow K^+ K^-) = (49.9 \pm 0.5) \times 10^{-2}$.

The first two branching fractions already include the branching ratio value of the $\phi \rightarrow K^+ K^-$ decay, as indicated by the final-state notation. The measured values of the cross section of the promptly produced charged D mesons in the LHCb acceptance are [4]

- $\sigma(pp \rightarrow D^+ X) = (834 \pm 2 \pm 78) \mu\text{b}$,
- $\sigma(pp \rightarrow D_s^+ X) = (353 \pm 9 \pm 76) \mu\text{b}$.

Please note that large systematic uncertainties are due to the absolute measurement of these values, so they largely, if not completely, cancel out in the ratio. Details on the source of the different uncertainties can be found in Ref. [4].

As already said above, the goal of this experiment is the search for a rare, still unseen, decay process, and therefore the analysis strategy should be developed as *blindly* as possible. There are no fully standard procedures or fixed rules to follow: the robustness of the blinding strategy ultimately relies on the discipline and integrity of the analyst performing the measurement. It is recommended not to inspect the invariant-mass region around the known value of the lepton mass, and not to examine the results of the signal-fraction fits, until the analysis strategy has been frozen and all validation checks have been completed.

Suggested procedures and measurements

1. Use the simulated samples to determine the invariant-mass shapes of the $\tau^+ \rightarrow \phi\mu^+$ signal decay, the semileptonic normalisation $D_s^+ \rightarrow \phi\mu^+\nu_\mu$ decay, and the two hadronic mis-identified backgrounds $D^+ \rightarrow \phi\pi^+$ and $D_s^+ \rightarrow \phi\pi^+$. All of them are reconstructed under the $\phi\mu$ mass hypothesis.
2. Build a 1D probability density function to model the distribution of the invariant mass $m(\phi\mu)$ observed in data. The model should include all the different components: $\tau^+ \rightarrow \phi\mu^+$, $D_s^+ \rightarrow \phi\mu^+\nu_\mu$, $D^+ \rightarrow \phi\pi^+$, $D_s^+ \rightarrow \phi\pi^+$ and combinatorial background.
3. Define a blinding strategy and perform a blinded fit to the data.
4. Validate the fit using pseudo-experiments generated from the nominal model. In particular, demonstrate that the maximum-likelihood estimators of the physics observables of interest exhibit Gaussian behaviour to a very good level of precision.
5. Construct a Feldman–Cousins confidence belt using pseudo-experiments generated at fixed values of $\mathcal{B}(\tau^+ \rightarrow \phi\mu^+)$. The construction should use the Gaussian approximation for the chosen estimator.
6. (*Optional*) Optimise the offline selection (p_T , ct , FD_T , ... of the τ meson) to maximise the expected sensitivity to the signal.
7. Measure $R_{\phi\pi}$ as a validation of the full analysis procedure.
8. Unblind the final result. Evaluate the significance of the observed signal with respect to the hypothesis $\mathcal{B}(\tau^+ \rightarrow \phi\mu^+) = 0$. Determine the observed $\mathcal{B}(\tau^+ \rightarrow \phi\mu^+)$ (*point estimate*), and report either the confidence interval or the upper limit at 90% CL (*interval estimate*).
9. The same as the previous point but for the measurement of R_τ .
10. Discuss the dominant sources of systematic uncertainty. Where possible, estimate their impact quantitatively.

Further suggestions

A useful model for the invariant mass of the $D_s^+ \rightarrow \phi\mu^+\nu_\mu$ decay candidates is the ARGUS probability density function, defined as

$$p(m; m_0, c, p) = \begin{cases} \frac{1}{N} m \left(1 - \left(\frac{m}{m_0} \right)^2 \right)^p \exp \left[c \left(1 - \left(\frac{m}{m_0} \right)^2 \right) \right] & \text{if } m < m_0, \\ 0 & \text{if } m \geq m_0. \end{cases}$$

The normalization factor N is defined over the fitted mass range $[m_L, m_H]$ as

$$N = \frac{m_0^2}{2c(-c)^p} (d_L - d_H),$$

where

$$d_{L(H)} = \Gamma(1+p) \cdot Q \left(1+p, c \left[-1 + \left(\frac{m_{L,H}}{m_0} \right)^2 \right] \right)$$

The terms d_L and d_H are expressed as a function of the Euler gamma function $\Gamma(z)$ and of the regularized upper incomplete gamma function $Q(a, z)$. This parametrisation is used with $c < 0$ and $p \geq 0$. This ARGUS function has support only below the endpoint m_0 and is set to zero for $m \geq m_0$.

```

1 inline double pdf_Argus(double *x, double *par)
2 {
3     const double m = x[0];
4     const double m0 = par[0];
5     const double c = par[1];
6     const double p = par[2];
7
8     if (m >= m0) {
9         return 0.0;
10    }
11
12    const double xL = 1.0 - (L_taumass_range/m0)*(L_taumass_range/m0);
13    const double xH = 1.0 - (H_taumass_range/m0)*(H_taumass_range/m0);
14
15    const double gammaA = ROOT::Math::tgamma(1.0 + p);
16    const double dL = gammaA * ROOT::Math::inc_gamma_c(1.0 + p, -c*xL);
17    const double dH = gammaA * ROOT::Math::inc_gamma_c(1.0 + p, -c*xH);
18    const double norm = (m0*m0)/(2.0*c*std::pow(-c,p)) * (dL - dH);
19
20    const double u = 1.0 - (m/m0)*(m/m0);
21    return m * std::pow(u,p) * std::exp(c*u) / norm;
22 }

```

Listing 1: C++ implementation of the normalized ARGUS probability-density function used for the semileptonic component.

The following minimal ROOT macro illustrates the basic structure of a pseudo-experiment loop. A toy dataset is generated from a known signal-plus-background model written in terms of component fractions:

$$P(m) = N_{\text{tot}} \Delta m [f_{\text{sig}} G(m) + (1 - f_{\text{sig}}) E(m)],$$

where $G(m)$ is a normalised gaussian and $E(m)$ is a normalised exponential function.

```

1 void test_pseudoexperiments() {
2     gRandom->SetSeed(12345);
3
4     const double mMin = 1.60;
5     const double mMax = 2.10;
6     const int nBins = 50;
7     const double binWidth = (mMax - mMin) / nBins;
8
9     const int nToys = 1000;
10
11    const double nTotTrue = 500.0;
12    const double fSigTrue = 0.5;
13    const double meanTrue = 1.78;
14    const double sigmaTrue = 0.020;
15    const double lambdaTrue = -4.0;
16
17    const char* modelFormula = "[0]*[5]*([1]*TMath::Gaus(x,[2],[3],1) + (1.0-[1])*([4]*TMath::Exp
    ([4]*(x-1.60))/(TMath::Exp([4]*(2.10-1.60))-1.0))";
18
19    // Known model used to generate the pseudo-experiments.
20    // TF1::GetRandom uses only the shape of the function.
21    TF1 generatorModel("generatorModel", modelFormula, mMin, mMax);
22    generatorModel.SetParameters(nTotTrue, fSigTrue, meanTrue, sigmaTrue, lambdaTrue, binWidth);
23
24    TH1D hfSigFit("hfSigFit", "Fitted signal yield;#hat{f}_{sig};Pseudo-experiments", 50, 0.0,
    1.0);
25
26    for (int iToy = 0; iToy < nToys; ++iToys) {
27
28        TH1D hToys("hToys", "Toy dataset;M;Candidates", nBins, mMin, mMax);
29
30        // const int nObs = gRandom->Poisson(nTotTrue); // Extended toy generation.
31        const int nObs = nTotTrue;
32
33        for (int i = 0; i < nObs; ++i) {
34            const double m = generatorModel.GetRandom(mMin, mMax);
35            hToys.Fill(m);
36        }
37
38        // Fit model:
39        // par[0] = Ntot
40        // par[1] = fsig
41        // par[2] = Gaussian mean
42        // par[3] = Gaussian width
43        // par[4] = exponential slope

```

```

44 // par[5] = bin width, fixed
45 TF1 fitModel("fitModel", modelFormula, mMin, mMax);
46 fitModel.SetParameters(nObs, fSigTrue, meanTrue, sigmaTrue, lambdaTrue, binWidth);
47 fitModel.SetParNames("Ntot", "fsig", "mean", "sigma", "lambda", "binWidth");
48 fitModel.FixParameter(0, nObs); // or comment to let it float
49 fitModel.FixParameter(5, binWidth);
50
51 hToy.Fit(&fitModel, "LI");
52
53 const double nTotFit = fitModel.GetParameter(0);
54 const double fSigFit = fitModel.GetParameter(1);
55 const double nSigFit = fSigFit * nTotFit;
56
57 hfSigFit.Fill(fSigFit);
58 }
59
60 TCanvas c("c", "Pseudo-experiment result", 800, 600);
61 hfSigFit.Draw("HIST");
62 c.SaveAs("toy_fitted_signal_yield.pdf");
63 }

```

Listing 2: Minimal ROOT example of a pseudo-experiment loop. Toy datasets are generated from a known Gaussian-plus-exponential model, fitted with a fraction-based model, and the fitted signal yield is stored.

References

- [1] Belle Collaboration, *Search for Lepton-Flavor-Violating tau Decays into a Lepton and a Vector Meson*, arXiv:1101.0755 [hep-ex] Phys.Lett.B 699 (2011) 251-257
- [2] D. Riccardi, *Search for the lepton-flavour violating decay $\tau \rightarrow \mu\phi$ at the LHCb experiment*, CERN-THESIS-2022-221
- [3] S. Navas et al. (Particle Data Group), Phys.Rev.D 110, 030001 (2024) and 2025 update. See also <https://pdg.lbl.gov/>.
- [4] LHCb Collaboration, *Measurements of prompt charm production cross-sections in pp collisions at $\sqrt{s} = 13$ TeV*, JHEP03(2016)159; JHEP09(2016)013; JHEP05(2017)074, arXiv:1510.01707 [hep-ex]
- [5] R. Aaij et al. (LHCb collaboration), LHCb Detector Performance, Int. J. Mod. Phys. A 30, 1530022 (2015). arXiv:1412.6352 [hep-ex].